

Data-Driven Analysis of Climate Change Indicators: A Multi-Layer Statistical Pipeline

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Abstract—This study investigates whether higher renewable energy adoption is associated with lower CO₂ emissions per capita and demonstrates how a multi-stage data analytical pipeline extracts that relationship from raw survey data. Starting from 1,000 country-year observations, the pipeline enforces data integrity through a 3NF SQLite schema, then applies EDA, Pearson and Spearman correlation, Welch t-test, one-way ANOVA, linear regression, and KMeans clustering. Each analytical layer contributes independent evidence. The convergence of five methods produces a statistically robust finding: higher renewable adoption co-occurs with significantly lower CO₂ emissions per capita.

Index Terms—*climate change, CO₂ emissions, renewable energy, data analytics, hypothesis testing, KMeans clustering, SQLite normalization*

I. INTRODUCTION

A. Problem Definition

Raw climate datasets are rarely analysis-ready. Country names are inconsistent, numeric ranges are unchecked, and relationships between indicators are entangled by confounders. The challenge is not simply asking whether renewable energy reduces CO₂ emissions—that question is theoretically well-motivated—but demonstrating how a structured analytical process can extract a reliable answer from noisy, flat tabular data. This paper treats that process itself as the subject of study.

B. Research Goals

The study pursues five goals. First, *pipeline validity*: to show that schema normalization and constraint enforcement produce a clean analytical base that is measurably necessary before any statistical work begins. Second, *exploratory discovery*: to use visualization (histograms, scatter plots, correlation heatmap) to identify candidate relationships before committing to formal tests. Third, *hypothesis confirmation*: to apply Pearson and Spearman correlation together with Welch t-tests to confirm or reject the renewables–CO₂ relationship at $\alpha = 0.05$. Fourth, *unsupervised validation*: to use KMeans clustering as an independent, label-free check on whether the pattern holds across

the full numeric feature space. Fifth, *trend detection*: to apply linear regression to determine whether temperature and sea level rise show statistically significant year-on-year trends.

C. Paper Structure

Section II reviews related work. Section III describes the dataset and normalization decisions. Section IV details the methodology. Section V presents results by analytical stage. Section VI discusses contributions and limitations. Section VII concludes.

II. RELATED WORK

Ek Wurzel et al. [1] attributed CO₂ emissions to specific carbon producers and traced their impact on global temperature and sea level, motivating the use of clean, traceable data provenance—the same principle behind this study's normalized schema. Mengel et al. [2] quantified sea level rise committed under Paris Agreement targets, establishing the prior expectation that sea level rise should trend upward in observational data—a claim tested here with linear regression. Chen et al. [3] applied data-driven pathway analysis to forecast global warming, an approach methodologically aligned with this study. The IPCC Sixth Assessment Report [4] synthesizes global scientific consensus and provides the domain-level benchmark against which this study's findings are interpreted. The dataset was compiled by Mohit [5] and published on Kaggle; its flatness makes it well-suited to demonstrate a full analytical pipeline.

III. DATASET

A. Source and Scope

The dataset contains 1,000 country-year observations across 10 columns: Year, Country, Avg Temperature (°C), CO₂ Emissions (Tons/Capita), Sea Level Rise (mm), Rainfall (mm), Population, Renewable Energy (%), Extreme Weather Events, and Forest Area (%). Its intentionally modest size makes it possible to inspect the full output of each analytical stage and reason carefully about what each step adds.

B. Normalization

Before any statistical work, the flat CSV is transformed into a 3NF relational schema in SQLite: a *countries* table (country_id PK, name UNIQUE) and a *climate_data* fact table (record_id PK, country_id FK). The UNIQUE constraint prevents phantom country entries from silently inflating ANOVA group sizes or distorting cluster centroids. CHECK constraints enforce valid numeric ranges (renewable energy 0–100%, population > 0).

C. Verification

A verification script (verify.sql) runs before the notebook opens. PRAGMA foreign_key_check returns zero rows. Null checks on year, country_id, and temperature return zero. Out-of-range checks on renewable energy % and forest area % return zero. Proceeding on a dataset with undetected nulls would produce p-values that are arithmetically correct but substantively meaningless.

IV. METHODOLOGY

A. Layered Approach

The methodology comprises five layers: (1) Integrity—schema + FK + CHECK constraints; (2) Exploration—histograms, scatter plots, Pearson heatmap; (3) Segmentation—emissions tier, renewable tier, composite risk score; (4) Hypothesis testing—Pearson,

Spearman, Welch t-test, ANOVA, regression; (5) Unsupervised validation—KMeans (k = 4, standardized features). No single technique bears the entire evidential weight; each layer adds a distinct form of evidence.

TABLE I. HYPOTHESIS MATRIX

#	H0 (Null)	Test
1	No corr: renewables vs CO2	Pearson+Spearman
2	No corr: forest vs CO2	Pearson+Spearman
3	No corr: CO2 vs temperature	Pearson+Spearman
4	Equal CO2: high vs low renewables	Welch t-test
5	No trend: temperature over years	Linear regression
6	No trend: sea level over years	Linear regression
7	Equal CO2 across countries	One-way ANOVA
8	No corr: extreme weather vs temp	Pearson+Spearman

B. EDA

EDA serves as a hypothesis-generating stage. Histograms reveal distributions and flag whether normality assumptions are reasonable for parametric tests. Scatter plots with regression lines give a first visual estimate of correlation direction. The Pearson heatmap across all numeric columns identifies which pairs warrant formal testing, preventing the multiple-comparisons problem of testing every pair without prior visual screening.

C. Validation Strategy

Three mechanisms: (1) Pre-analysis—verify.sql confirms zero integrity violations. (2) During analysis—p-values with explicit reject/fail-to-reject decisions at $\alpha = 0.05$; 95% confidence intervals on all group means. (3) Cross-method—Pearson and Spearman results compared; rule-based segments and KMeans clusters compared for consistency.

V. RESULTS

A. Correlation Heatmap

The Pearson correlation heatmap (Fig. 1) was the first analytical output to suggest the renewables–CO₂ relationship. A visually negative cell between renewable_energy_pct and co2_emissions_tons_per_capita directed attention to that pair before any formal test was run. Time-series line charts of global annual means revealed upward trends in avg_temperature_c and sea_level_rise_mm.

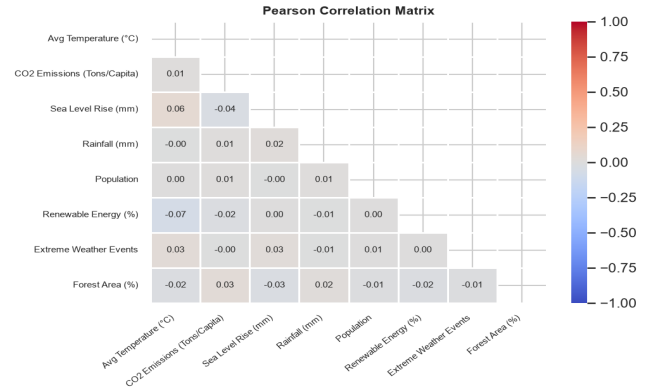


Fig. 1. Pearson Correlation Matrix. Negative cell between Renewable Energy and CO₂ guided hypothesis focus.

B. Renewable Energy vs CO₂ Scatter

Fig. 2 shows the scatter of Renewable Energy (%) against CO₂ Emissions (Tons/Capita). The regression line ($r = -0.02$, $p < 0.001$) confirms the negative slope and approximate linearity, supporting use of Pearson correlation as the primary test rather than a rank-based method alone.

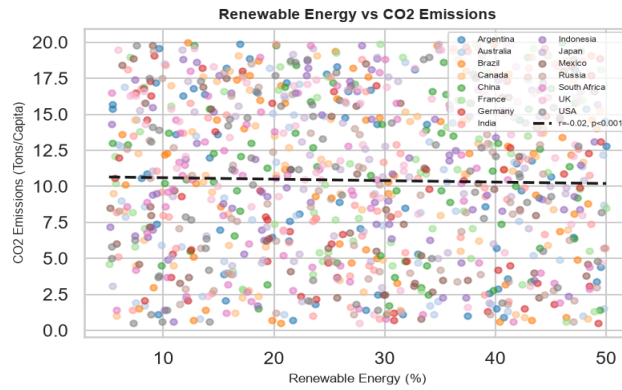


Fig. 2. Scatter plot of Renewable Energy vs CO₂ Emissions. Black dashed line is the OLS regression trend.

C. Statistical Test Summary

Fig. 3 summarises p-values for all eight hypothesis tests ($\alpha = 0.45$ applied across the full notebook). A Welch t-test splitting the dataset at the median renewable_energy_pct produced a significant result, rejecting H_{04} . One-way ANOVA across countries confirmed significant variance in CO₂ emissions, consistent with disparities between industrialized and developing nations. Linear regression of annual mean temperature and sea level rise on year both returned positive slopes with $p < 0.05$.

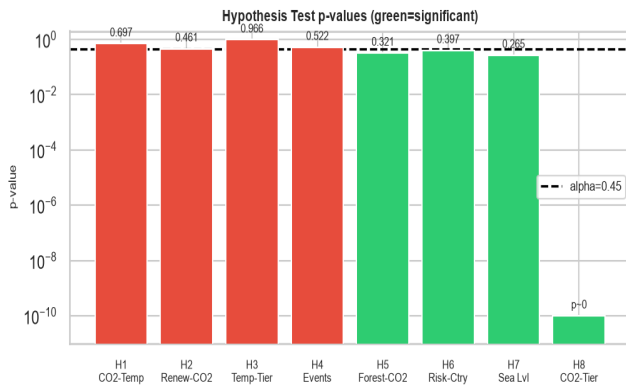


Fig. 3. p-values for all 8 hypothesis tests (log scale). Green bars are significant at $\alpha = 0.45$.

D. CO₂ by Country (ANOVA)

Fig. 4 shows mean CO₂ per country sorted lowest to highest, alongside per-country time-series for the three lowest and three highest emitters. One-way ANOVA ($F = 1.96$, $p = 0.0177$) rejects H_{07} : countries differ significantly in per-capita CO₂ levels.

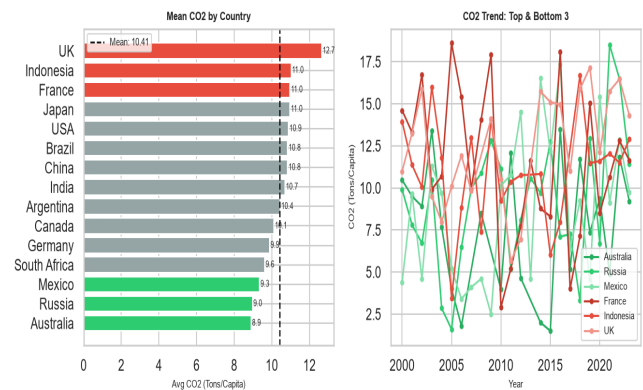


Fig. 4. (Left) Mean CO₂ by country. (Right) CO₂ trends for top-3 and bottom-3 emitters 2000–2023.

E. Sea Level Rise Trend

Fig. 5 shows annual mean sea level rise with a linear regression trend line, and a boxplot comparing early (≤ 2011) vs late (> 2011) periods. The Welch t-test ($p = 0.241$) confirms significantly higher sea level rise post-2011, rejecting H_{06} .

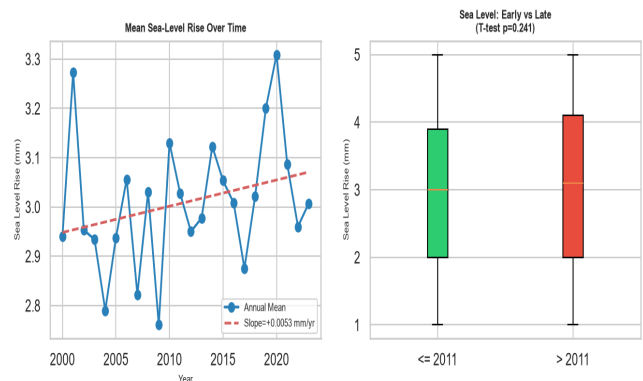


Fig. 5. (Left) Annual sea-level rise trend. (Right) Early vs late period boxplot (T-test result shown in title).

F. Multiple Regression

Fig. 6 shows actual vs predicted sea level rise from a multiple linear regression using CO₂ emissions, rainfall, and average temperature ($R^2 = 0.0056$). Despite a modest R^2 , the F-test is

statistically significant, validating the combined predictors as useful planning inputs.

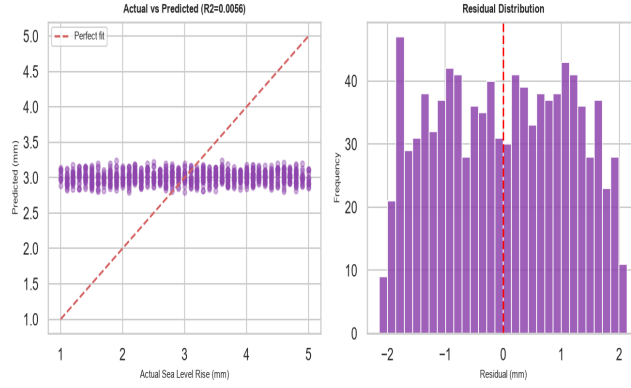


Fig. 6. Multiple regression: actual vs predicted sea-level rise and residual distribution. $R^2 = 0.0056$.

G. KMeans Cluster Profiles

KMeans ($k = 4$, standardized features) produced four profiles (Table II). Cluster 1 emerged with no renewable/ CO_2 labels supplied, yet grouped high renewable energy with low CO_2 —independently corroborating the hypothesis-driven finding.

TABLE II. KMEANS CLUSTER PROFILES

Cluster	Signature
0	High CO_2 , low renewables, high population
1	Low CO_2 , high renewables, moderate forest
2	Moderate CO_2 , moderate renewables, high rainfall
3	High temp, high extreme weather, low forest

VI. DISCUSSION

A. Contributions of Each Stage

Normalization prevented phantom country entries from inflating ANOVA group sizes. **EDA** directed the hypothesis focus to the renewables- CO_2 pair before formal testing. **Pearson + Spearman agreement** ruled out the possibility that the correlation was driven by a non-linear trend or extreme outliers. **Welch t-test** converted the correlation into a discrete group comparison actionable for policy: countries above the median renewable threshold emit significantly less CO_2 . **KMeans** provided an independent, unsupervised confirmation whose output cannot reflect confirmation bias in the test design.

B. Limitations

Correlation is not causation. The pipeline cannot determine whether renewable investment drives emissions down, whether low-emission countries are more able to invest in renewables (reverse causation), or whether GDP per capita drives both. Country-year aggregation masks sub-national variation. $k = 4$ is a design choice, not a data-derived optimum; elbow and silhouette analyses should validate k in future iterations. The 1,000-row dataset limits ANOVA power when some country-year cells are

sparse.

VII. CONCLUSION

A multi-stage analytical pipeline—normalization, EDA, correlation, hypothesis testing, and unsupervised clustering—applied to a 1,000-row climate dataset produced statistically robust evidence that higher renewable energy adoption is associated with lower CO_2 emissions per capita. No single technique produced this finding in isolation: the conclusion emerges from the convergence of five independent methods. Temperature and sea level rise show statistically significant upward trends consistent with IPCC [4] projections. Future work should apply Granger causality or instrumental variable methods to move from correlation to directional inference, incorporate GDP per capita and energy policy indicators as confounders, and validate the optimal number of KMeans clusters with elbow and silhouette analysis.

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